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Science Fair Final Report

Fuel For Thought

Eighty-seven octane, eighty-nine octane, ninety-three octane, ninety octane No-Ethanol, eighty-eight octane E15, and E85. These are often the different fuels you may run into at the gas station. With that being said, have you ever wondered what the difference is, other than the cost? And if you do know the difference, is there actually any advantage to the more expensive fuel? Lastly, what is the most fair way to determine which fuel is the most efficient? These were just a few of the questions we wanted to answer with our science fair project.

In order to understand our project, it is helpful to know the differences between each fuel if you don't already. Eighty-seven octane is the most common fuel, and typically the cheapest at your local gas station. The number eighty-seven is there to represent the ability, or lack thereof, for the fuel to 'detonate'. For example, a fuel that measures one-hundred octane would require more compression to be ignited efficiently than the typical eighty-seven octane. Furthermore, eighty-seven octane in Ohio and other states consists of 10% ethanol to save costs and be more environmentally friendly (Larry Carley). Ethanol is a type of alcohol added to traditional gasoline that is derived from corn and other organic starches. Eighty-nine octane, similar to eighty-seven, contains 10% ethanol at most places (Carley). Eighty-nine octane fuel is typically more expensive than eighty-seven due to it being advertised as a "higher quality" fuel with extra additives and detergents to help clean engine components. Ninety-three is the most expensive fuel we tested other than the no-ethanol gasoline. Similar to eighty-nine, it has increased

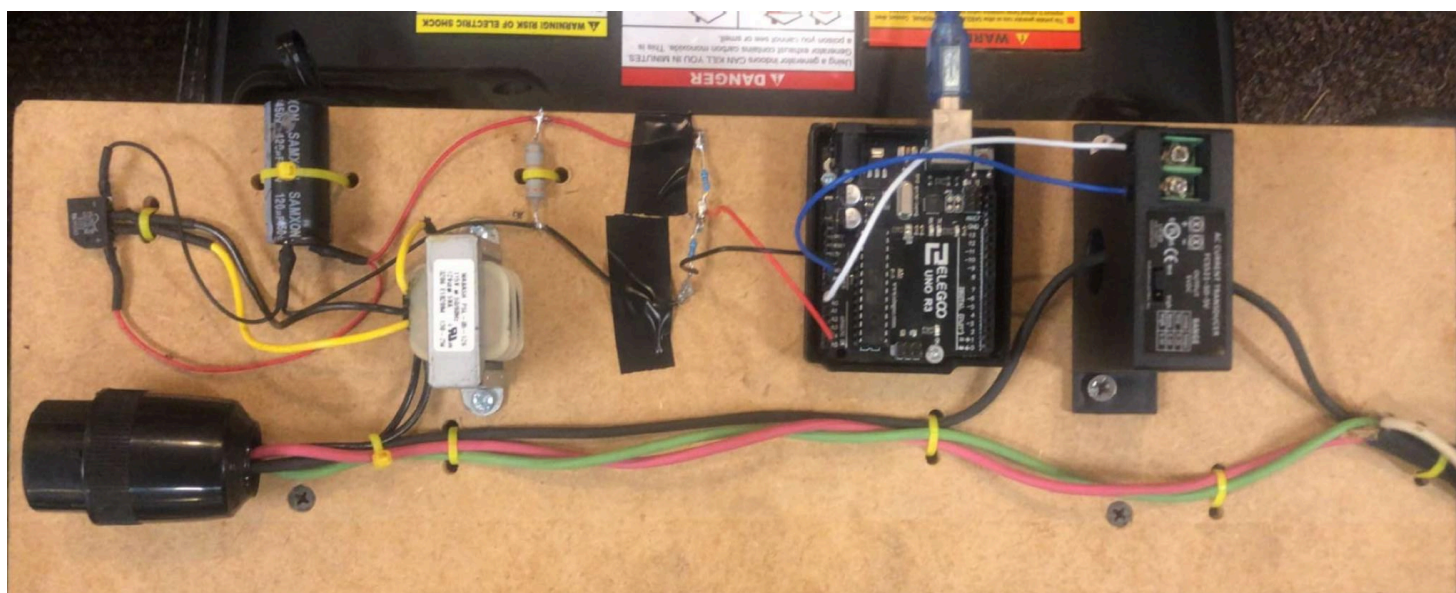
additives and is 10% ethanol (Carley). However, it is often advertised as even more premium than ninety-one octane.

The next type of fuel is ninety octane no-ethanol which is similar to older types of fuel. Not to be confused with leaded gasoline which contains tetraethyl-lead as an additive. This fuel is pure modern gasoline, without ethanol, that can be found at some specialty gas-stations and is generally used for classic cars. The reason it is commonly used for classic cars is because it does not contain ethanol. Ethanol can be damaging to older vehicles because their seals and gaskets were not made to withstand ethanol in fuel. Ethanol can cause older rubber gaskets and seals to swell which can cause a variety of problems in the fuel systems of old cars, especially ones that are carbureted. However this is not a problem in modern cars with synthetic seals that are made for the presence of ethanol. Eighty-eight octane E15 is a little bit different from the rest, as it contains eighty-five percent gasoline and fifteen percent ethanol. This increased ethanol percentage makes it the second cheapest fuel on our list.

Lastly, E85 is by far the most unique of this list. Its composition is much different than the rest, being about 85% ethanol fuel and 15% gasoline which varies with the weather (Carley). E85 is typically much rougher on engines due to the alcohol, ethanol, washing oil from the cylinder walls, and accelerating wear. Further, it is known to have a much harder time than normal gasoline igniting when cold. For this reason, its composition is heavily varied from 50% to 85% throughout the months in colder climates. Meaning its octane can also vary from 100-115 depending on the current blend. Something that really makes E85 stand out from the rest is its energy density. The energy density of a fuel is how much energy is contained in a volume of the fuel. In this case, E85 is typically 23-27 MJ per liter once again depending on the mix ("Energy Content of Biofuel"). An advantage of E85 is its ability to produce more power than most normal

gasoline types. E85 is typically used for race cars because of its higher octane, more forgiving for higher compression ratios or artificial aspiration, such as turbochargers and superchargers. The main attraction of E85 to the average person is its environmental benefits. Ethanol is a renewable fuel type which can be made using many different types of plants and vegetables. Additionally, pure ethanol when burned produces about two kilograms of CO₂ per kilogram burned, compared to the three kilograms of CO₂ that gasoline produces (“Energy Content of Biofuel”). This may be misleading as E85 is about 27% less energy dense, therefore more must be burned for the same power. Per kilogram of CO₂ produced by pure ethanol, it produces about 12.25-14.03 megajoules of energy which is still about 0.5-1 megajoules better than typical gasoline would produce (“Energy Content of Biofuel”).

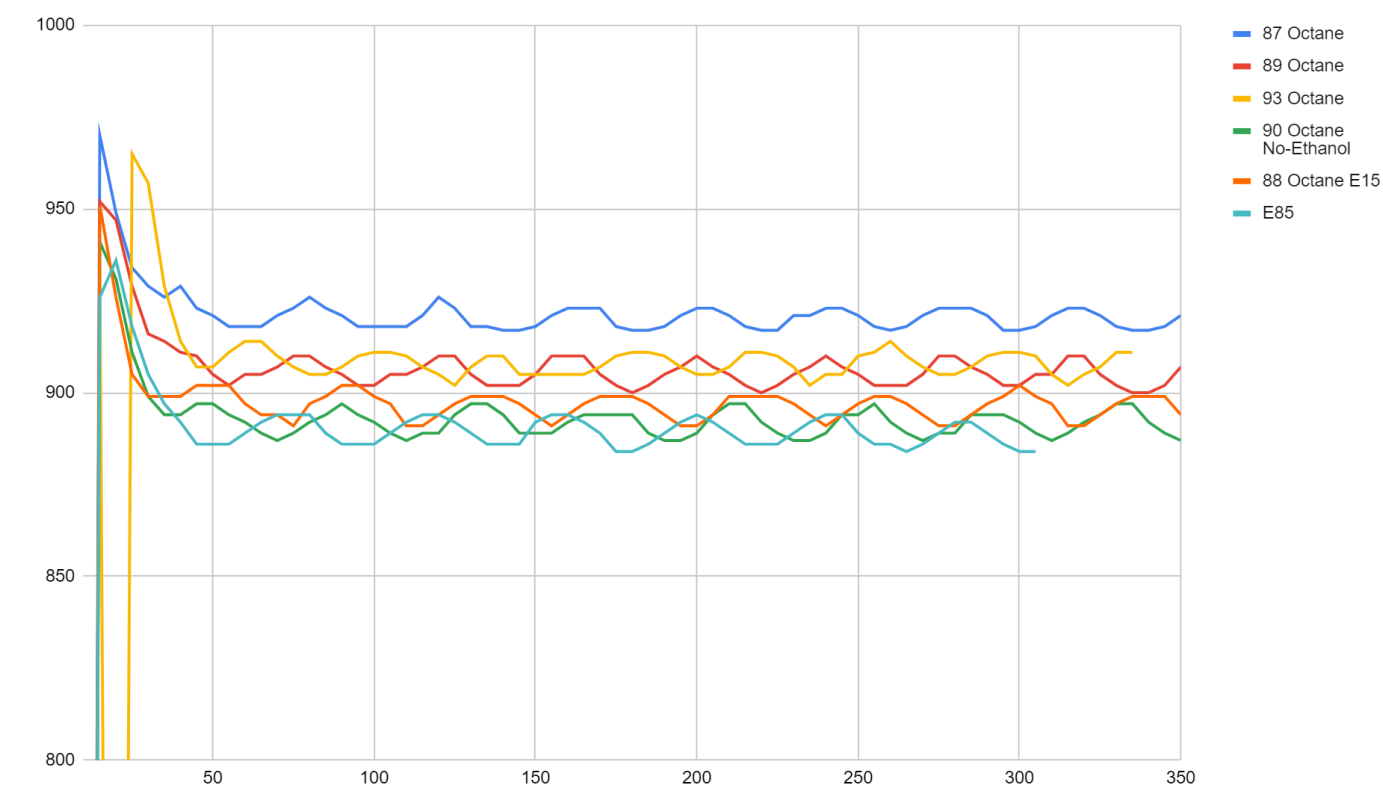
Now that we have covered the difference of the different types of fuels, it is time to test the fuels, answer our questions, and assess our hypothesis. To execute our project, some of the main things we were going to need were a generator, an arduino with measuring equipment, a space heater, and the six different types of fuel that we would be testing and comparing. We will then compile our data to different units such as run time and kilojoules of energy produced. Next, we will organize the data collected by the arduino in a spreadsheet and graph the data points collected. For the results of the test, we are expecting all three 10% ethanol fuels to perform similarly (eighty-seven, eighty-nine, and ninety-one) since they should have very similar energy density. We expect eighty-eight octane to perform slightly worse than the 10% ethanol fuels and E85 to perform the worst due to the decreased energy density of ethanol. We believe that ninety-octane will perform slightly better than any of the other fuel types due to it having the highest energy density.



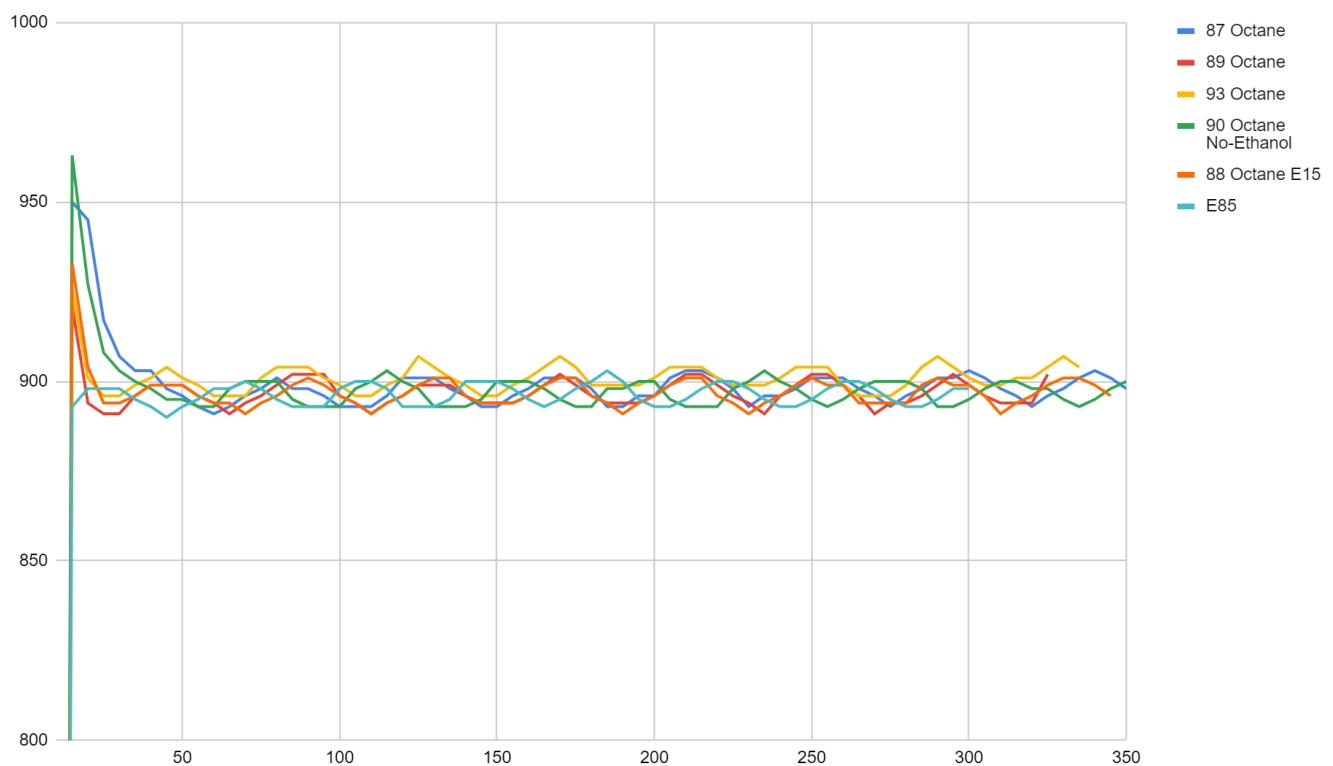




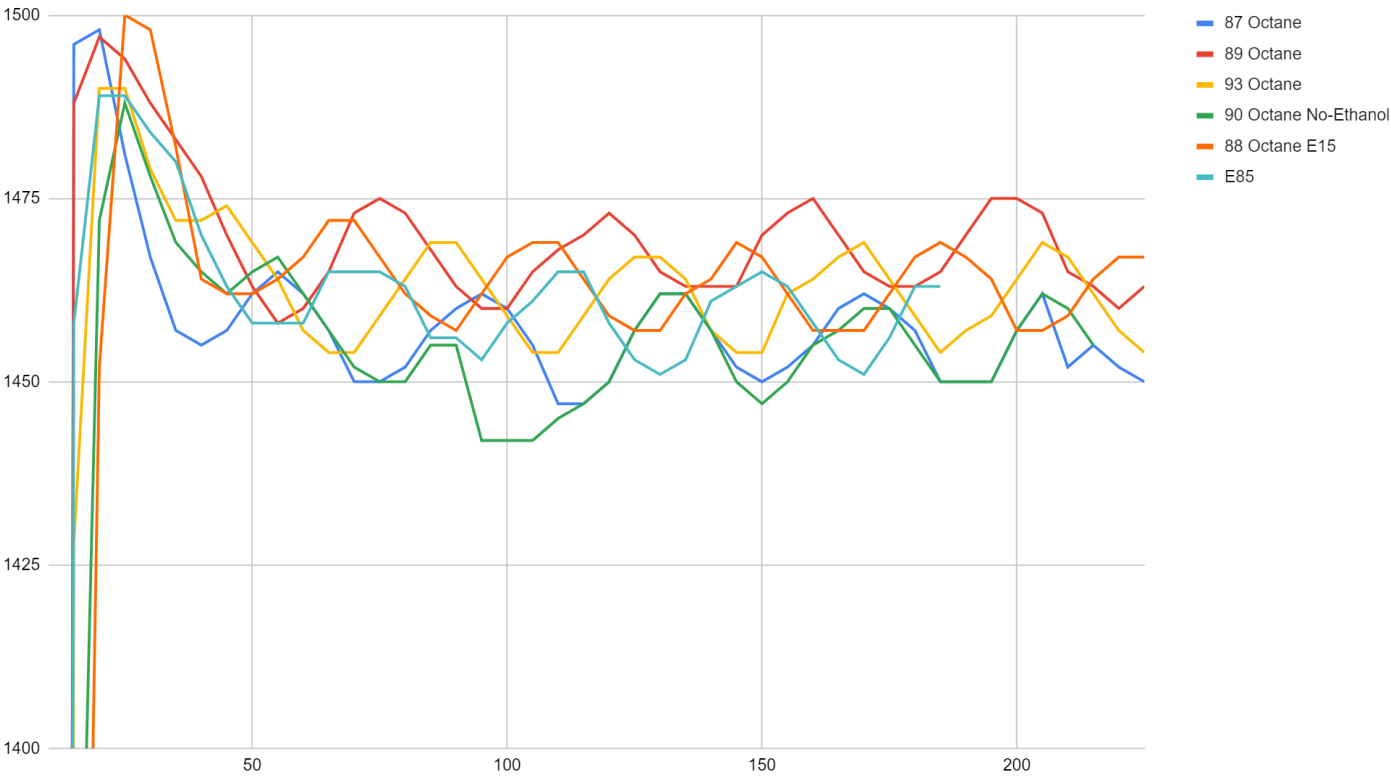
1st Trials Watts vs Time (seconds) - Credit: Collin Snider and Eli Voisard



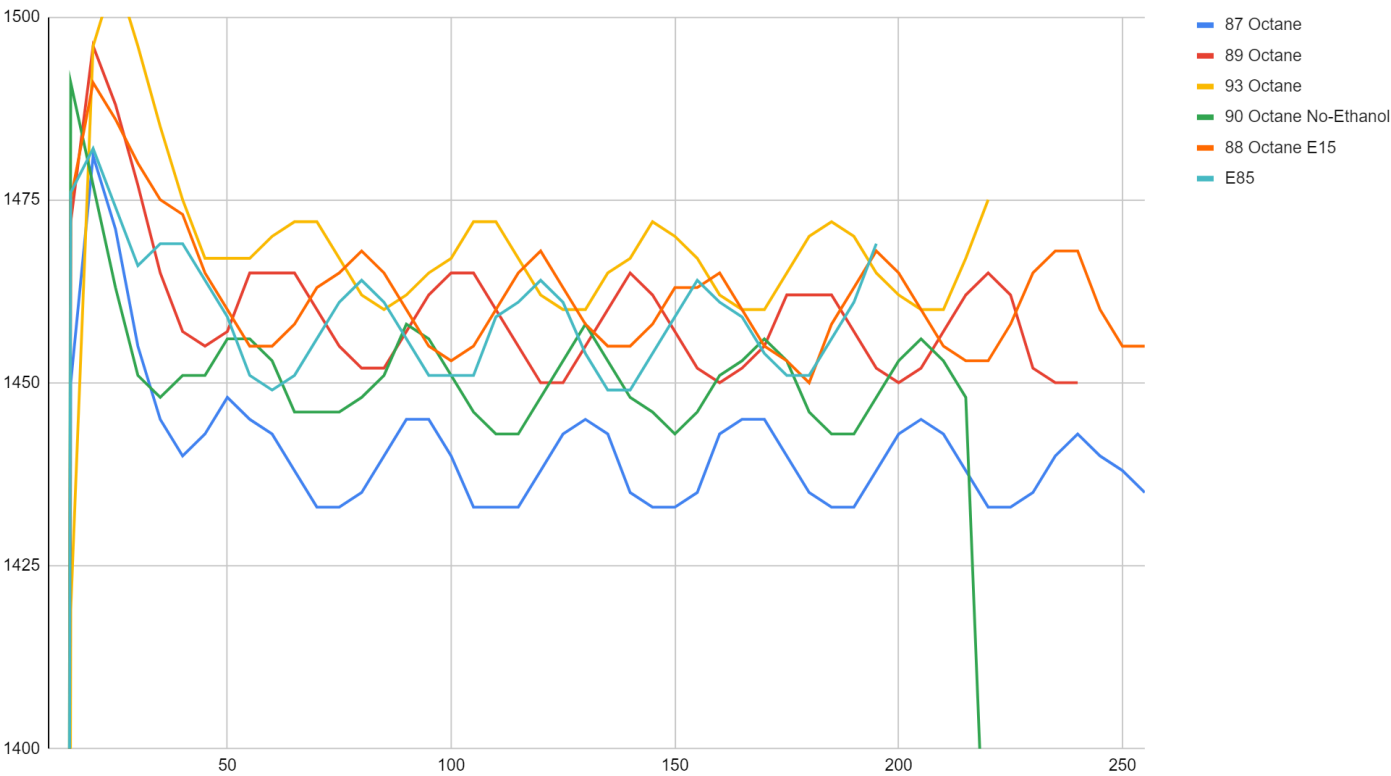
2nd Trials Watts vs Time (seconds) - Credit: Collin Snider and Eli Voisard



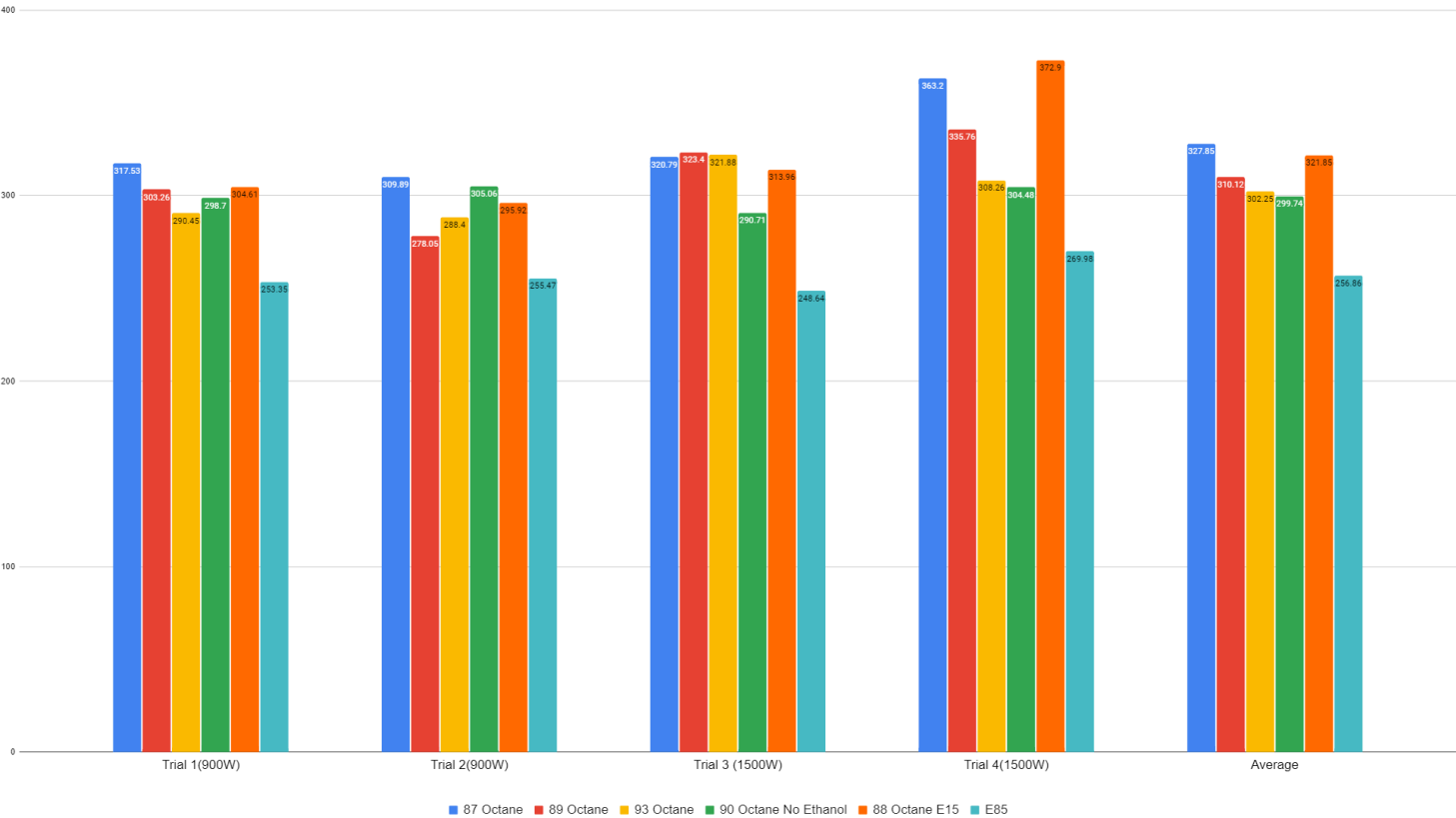
3rd Trials Watts vs Time (seconds) - Credit: Collin Snider and Eli Voisard



4th Trials Watts vs Time (seconds) - Credit: Collin Snider and Eli Voisard



Experimental KiloJoules of Energy in 70 mL of Each Fuel - Credit: Collin Snider and Eli Voisard



After compiling the data collected from our experiment, we noticed some interesting findings. The main unit of which we will be comparing the different fuel types is by kilojoules, which is a combination of each data point collected, amps, voltage and time. We calculated this by first converting the voltage and amperage to watts, with assuming a power factor of 1 for the heating element, and then watts to watt hours by adding time to the equation. Finally, we converted the watt hours to kilojoules as it is a more accepted unit of energy in the sciences.

During trials 1 and 2, we had set the space heater to produce a load of about 900W (a little more than 1HP), this would be consistent across all fuels in these trials. Under the lighter load, 87 octane, had excelled over the rest, producing 313.17 kJ of energy. Surprisingly, 88 octane E15 performed very similarly producing an average of, and 300.27kJ. This number was closely matched by the much more expensive ethanol-free gas. Although we had expected 89 octane, 93 octane, to match or exceed 87 octane, all of these had slightly underperformed our hypothesis. Both of these fuels ended up netting about 290 kJ from 70mL. Lastly as we had predicted, E85 was far behind in results with only an average of 254.41 kJ produced in these trials.

Under higher engine loads of 1500W (2 mechanical HP), the results changed slightly. Under these higher loads, we can assume that the small engine we were using was much closer to its peak efficiency as most fuel types saw a great improvement. In these trials, to our amazement, 88 octane E15 performed better than any other yielding, 343.43 kJ of energy on average. This type of fuel was the second cheapest, often about 30 cents cheaper than unleaded 87, only more expensive than E85 and this result was very impressive to us. 87 octane was close to follow with a mean of 342.00 kJ netted between the tests. The 89 and 93 octane fuels both saw improvements under these conditions as well and they produced about 329.58 kJ and 315.07 kJ of energy respectively. Interestingly we did not see a large improvement in the 90 octane no-ethanol as we did the other fuels. It produced an average of 297.60 kJ, 4 kJ less than 301.88 recorded under a

lighter load. We suspect this is because the 100% gasoline burns slower than ethanol, meaning the piston “outruns” the explosion at higher RPMs like we saw in this test. Similar to the 90 octane, E85 did not see a vast improvement in energy efficiency. 259.31 kJ were measured to be the average which is only about 5 kJ better than the previous trials. Unlike the 100% gasoline, we suspect this to be a limitation of the testing equipment. E85 needs about 35% more fuel by volume of which the carburetor on the generator likely could not fully flow leading to higher EGTs and decreased efficiency.

As a result of four trials that included six different gasoline/ fuels being tested, and over 1,600 data points collected, we concluded that eighty-eight octane E15 was the most efficient fuel in producing energy kilojoules at different loads of power while also factoring in cost. Eighty-eight octane E15 produced the second highest amount of energy per 70 mL of liquid, second to eighty-seven octane. Eighty-seven octane produced an average energy value of 327.85 KJ. Eighty-eight octane E15 produced an average energy value of 321.85 KJ. Specifically on the 1500 watt trials, the energy produced by eighty-eight octane E15 was very similar to that of the eighty-seven octane. This proves that eighty-eight octane E15 is equally efficient in higher loads. Although the eighty-seven octane produced six more kilojoules of energy on average, we determined that the eighty-eight octane E15 was better because it was around nine cents cheaper per gallon. We also inferred that if we were to do more trials, the difference in energy between eighty-seven octane and eighty-eight octane E15 would become smaller. We also concluded that there is no reason to get the premium gas unless your car recommends it, because eighty-seven octane, the regular gas, produces the highest amount of kilojoules. Additionally, we acknowledged that E85 is the cheapest fuel but will likely be more expensive in the long run because of damage it can do to parts of your car. With all that being said, our tests led us to believe that eighty-eight octane E15 was the most efficient fuel because it produced a higher

amount of energy, works equal to or slightly better under heavier loads, and will cost less than other fuels in the long run.

Works Cited

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Abstract

Fuel For Thought

The purpose of our experiment was so we could determine what the differences between various fuels were in their performance. The question we asked was, what type of gasoline/ fuel can produce the most kilojoules of energy with 70 mL of liquid under different loads of power? We hypothesized that eighty-eight octane will perform slightly worse than the 10% ethanol fuels and E85 to perform the worst due to the decreased energy density of ethanol. We believe that ninety-octane will perform slightly better than any of the other fuel types due to it having the highest energy density. We conducted the experiment by adding 70mL of the selected fuel into the generator. We then started the generator, space heater, arduino, and computer code. Next, we organized the data from the arduino into a spreadsheet and repeated for each trial. These tests led us to believe that ninety octane is the most efficient out of the fuels we tested. It also showed that the 10% ethanol fuels were very similar to each other. Also as expected, the eighty-eight octane and E85 were the worst performers. This led us to conclude that there is really no point to getting a more premium, and expensive fuel if your only concern is efficiency.